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Beyond altruism: why intention, not motivation, correlates with blood donation behavior among young Indian adults

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Abstract

Background The issue of blood donation is still a problem in India, since there is a disconnect between individuals' willingness to donate and their actual donation practices. Even though the role of altruism in donation motivations is well acknowledged, the relationship between it and intention as well as past behavior is yet poorly explored among young adults.

Objectives Research Objectives: To determine if altruistic motives had varying associations with donation intentions and self-reported donation behaviors, and if the association between donation intentions and self-reported donation behaviors was stronger than that of altruism.

Methods A cross-sectional online survey was conducted among adults aged 18–40 years. *Starting from an initial pool of 580 responses, a rigorous screening process was applied to ensure data quality and age compliance, resulting in a final analytical sample of 95 participants (84% exclusion rate).*

Results Altruistic motivation was associated with higher donation intention (OR = 3.62, 95% CI [1.36–9.68], p = .010), but was not independently associated with reported past donation behaviour after accounting for intention (OR = 0.96, 95% CI [0.30–3.08], p = .949). Higher donation intention was associated with higher reported past donation frequency (OR = 4.42, 95% CI [2.21–8.81], p < .001). Female participants had lower odds of reporting higher donation intention.

Conclusion Findings are consistent with a discrepancy between donation intention and reported donation behaviour. Given the cross-sectional design and substantial sample reduction, results should be interpreted cautiously as exploratory associations within this analytical sample.

Keywords Altruism, Blood donation, Intention - action gap, Donation behavior, Young adults, India, Ordinal logistic regression

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Introduction

Blood donation is an important element in the healthcare system across the globe but sustaining an adequate supply has always been an issue, especially in India where, despite having millions of young adults, there is still a notable shortfall in terms of voluntary blood donation [1–3]. While research has extensively covered donor behavior in various regions, recent studies in developing contexts, such as Afghanistan, continue to highlight that Knowledge, Attitude, and Practice (KAP) remain critical factors in determining safe blood donation rates among public university students [4]. Although conventional methods of motivating people have mainly centered around the appeal for their sense of moral duty, evidence in the literature demonstrates that there exists a gap between the two, referred to as the intention-behavior gap [3, 5]. Presently, existing literature shows that although there is a willingness and desire among people to donate, few act out those intentions. It can thus be inferred that while moral motivation plays a crucial role in making people develop intentions to donate blood, it cannot overcome certain obstacles at the time of action. Most of the research done in India on this topic so far has addressed awareness alone, with little information regarding the motivations behind closing that gap. Comprehending these processes is critical for the successful implementation of donor retention strategies [6–8].

In this research, attention will be paid to the differences between altruistic motive, donation intention, and the past experience of donations by young Indians. Unlike studies which take altruistic motive and the behavior of donation as the same process, the research will determine whether the altruistic motive leads to the intention, and if donation intention is closer to the frequency of past donations.

Literature review and theoretical framework

The current investigation is theoretically based on the Theory of Planned Behavior (TPB) [9]. According to TPB, intention is the closest predictor of behavior. Nevertheless, it has been suggested previously that intentions do not necessarily lead to actions, especially regarding prosocial behavior and health-related behaviors [7, 10, 11].

Earlier research has pointed out different motivational factors, but altruism, defined as selflessness or concern for others' welfare, emerged as one of the major reasons behind people's willingness to contribute initially. However, as pointed out by modern studies [7, 10], pure altruism might need Path associations in terms of concrete "humanization" factors or emotional influences before translating into real-life actions. The Indian case is limited to descriptive "Knowledge, Attitude, and Practice" (KAP) parameters. There is no information available on the connection between psychological motivation, intention, and real-world actions among the 18–40 age group population.

Recent work on humanization suggests that prosocial intention may be strengthened when beneficiaries are represented in emotionally salient and humanized ways. Borinca, McAuliffe, and Nightingale (2024) show that humanizing information can improve intergroup affect, partly by reducing anxiety and increasing positive affective responses. Although their work is not specific to blood donation, it supports the broader argument that abstract prosocial motivation may require a more concrete emotional connection to become behaviorally meaningful. In the present study, humanization is discussed as a theoretical extension rather than a measured variable [12].

Research objectives

Keeping the above considerations in view, this study is proposed to examine:

1. The association between altruism and intent to donate blood among young Indians.
2. Whether there is an association between donation intent and prior donation experience.
3. If there exists any significant relationship between altruism and behavior when donation intent is controlled.

Methodology

Study design

This cross-sectional survey employed an online survey instrument to explore the linkages between altruism, donation intentions, and prior donation behavior in young Indians. While such studies allow for investigation of linkages among variables within a set period, it does not enable causal inference or determination of cause-and-effect sequences.

Sample and participant selection

Respondents were recruited using non-randomized convenience sampling techniques via email and social media outlets.

Out of the 580 surveys conducted, 312 were removed from the study due to their non-conformity with the predetermined age limit of 18–40 years; 128 were also omitted since some key data variables were missing, while 45 were removed due to data quality issues, such as straight lining and less than 60-second completion time. The final sample size for data analysis thus stood at 95 cases. Complete case analysis was used without any imputation process. While this provided a way to ensure that all cases had complete data for the regression models, there is a risk of selection bias due to the high exclusion rate.

Data collection instrument

A multi-item measure of altruistic motive was administered by self-report based on scales from previous research on prosocial behavior and blood donation. Selected items were: "I derive a feeling of pleasure from helping other people," and "Even without anybody knowing of my contribution, I would still agree to donate." Ratings for the items were made using a five-point Likert scale, and the mean score was computed as the individual altruism score. The instrument demonstrated satisfactory reliability (Cronbach's $\alpha = 0.81$) in the current study population.

Statistical analysis

Analysis of the data was carried out via stats models in Python. The fact that the dependent variable is ordinal means that proportional odds ordinal logistic regression was employed.

Although being a first-time donor is more of a behavioral experience than a rate, there is theory behind the assumption of 'first-time donor' serving as a benchmark in an ordinal regression model [13]. As per the Donor Career Path theory, donation behavior is not only a measure of how many times a person donates but a process of becoming a committed donor that begins with the initiation stage and ends with habituation and maintenance. This means that the ordinal measure reflects not only the intensity of behavioral commitment but the evolution from a new to an established donor status.

Empirical support

In addition to the conceptual framework, the selection of the model is validated by diagnostic tests. The proportional odds assumption for the ordinal models was tested and satisfied through Brant's test ($p > .05$). This means that the association between the independent variables (intention, for instance) and the probability of being assigned to a 'higher' category is stable regardless of the thresholds in the scale.

There were two main models considered:

- Model 1 (Intention): Predicted by altruistic motivation, gender (Female = 1, Male = 0), and age group (25–40 = 1, 18–25 = 0).
- Model 2 (Behavior): Predicted by altruistic motivation, donation intention, gender, and age group.

Table 1 Frequency counts for ordinal outcome variables ($N = 95$)

Category	Intention to donate (future)	Past donation behaviour
1 (Lowest)	2 (Not at all)	46 (First-time donor)
2	24 (Rarely)	28 (Rarely)
3	40 (Occasionally)	20 (Occasionally)
4 (Highest)	29 (Regularly)	1 (Regularly)

Altruism was mean-centered before conducting the analysis. All the findings have been expressed using Odds Ratio (OR) values and the associated 95% confidence intervals. Proportional odds assumptions were tested using the Brant Test while the model fitness was evaluated by likelihood ratio Chi-Square and R^2 pseudo value. Due to the cross-sectional nature of the study, predictive analysis is difficult to make. However, considering both models, one can identify predictive relationships among the independent variables.

Since this is a cross-sectional study, any correlation found will not be able to establish causation. The order of events between the intention and behavior cannot be determined.

Because donation intention and past donation behaviour were measured at the same survey occasion, the analysis cannot establish temporal ordering between intention and behaviour. Past donation behaviour was treated as a self-reported indicator of prior behavioral engagement rather than as evidence that current intention led to later action.

Ethical considerations

This study received institutional approval from the Ethics Review Board. Informed consent was provided electronically from all subjects prior to data gathering. Anonymity and confidentiality were ensured during the process of storing data.

Results

Sample characteristics

A total of $N = 95$ cases passed the screening process out of the initial 580 survey attempts made. These were further screened, as presented in the STROBE flowchart in Fig. 3, through age criteria (18–40 years only) and missing responses in critical variables or poor quality (survey completion time below 60 s).

The majority of respondents were male (74.7%, $n = 71$) and between the ages of 18 and 25 years (78.9%, $n = 75$).

Table 1 presents the frequency distributions for the ordinal outcome variables. It is worth noting that although the respondents had high intentions to donate (mainly occasionally and regularly), they displayed low levels of behavior, with nearly half of them having previously donated blood only once before the study (48.4%).

Predictors of donation intention (Model 1)

An ordinal logistic regression was conducted to identify predictors of donation intention. The model suggests a good fit (likelihood ratio $\chi^2(3) = 21.60$, $p < .001$), with a pseudo-R^2 of 0.098.

As presented in Table 2, altruistic motivation was a statistically significant predictor of donation intention (OR = 3.62, 95% CI [1.36, 9.68], $p = .010$). Participants

Table 2 Ordinal logistic regression predicting donation intention

Predictor	Odds Ratio (OR)	95% Confidence Interval	*p*-value
Altruism (centered)	3.62	1.36–9.68	0.010
Gender (1 = female)	0.14	0.05–0.38	< 0.001
Age (1 = 25–40)	1.13	0.42–3.05	0.810

Model fit: Likelihood ratio χ^2 (3) = 21.60, $p < .001$; pseudo-R^2 = 0.098

Table 3 Ordinal logistic regression predicting past blood donation behaviour

Predictor	Odds Ratio (OR)	95% Confidence Interval	*p*-value
Intention (ordinal)	4.42	2.21–8.81	< 0.001
Altruism (centered)	0.96	0.30–3.08	0.949
Gender (1 = female)	2.51	0.82–7.71	0.108
Age (1 = 25–40)	1.52	0.54–4.28	0.426

Model fit: Likelihood ratio χ^2 (4) = 22.84, $p < .001$; pseudo-R^2 = 0.113

who scored higher on altruistic motivation had much stronger odds of reporting a higher level of intention to donate blood.

Gender was also found to be a significant predictor of donation intention. Female participants had much lower odds of reporting higher donation intention than male participants (OR = 0.14, 95% CI [0.05, 0.38], $p < .001$), after controlling for altruistic motivation. Age group was not a significant predictor of donation intention in this model (OR = 1.13, $p = .810$).

These findings suggest that altruistic motivation is associated with blood donation intention among young adults.

Associations with reported past donation behaviour (Model 2)

A second model evaluated predictors of past blood donation frequency. The model was statistically significant. The predictors in this model were altruistic motivation, donation intention, gender, and age. The model was significant (likelihood ratio χ^2 (4) = 22.84, $p < .001$), and it explained a pseudo R^2 of 0.113. The results of this model are shown in Table 3.

As presented in Table 3, Higher donation intention was strongly associated with higher self-reported past donation frequency within the ordinal regression model (OR = 4.42, 95% CI [2.21, 8.81], $p < .001$). Individuals who reported stronger intentions were much more likely to report stronger past frequencies of blood donation.

Conversely, altruistic motivation was not found to be significantly related to past donation behaviour after accounting for the effect of intention (OR = 0.96, 95% CI [0.30, 3.08], $p = .949$). Gender and age were also not statistically significant predictors of behaviour in this model.

These results suggest that altruistic motivation was associated with donation intention but was not independently associated with reported past donation frequency after intention was included in the model. This pattern is consistent with the study's expectation that intention would show a stronger association with reported behaviour than altruism.

Model-implied pattern of associations

Across the two ordinal models, altruistic motivation was associated with donation intention, and donation intention was associated with reported past donation frequency. Altruistic motivation was not independently associated with reported past donation frequency when intention was included in the model. Figure 1 presents this model-implied pattern of associations. Because the study is cross-sectional, the figure should be interpreted as a conceptual summary of observed associations rather than evidence of mediation or temporal ordering.

The solid arrows denote statistically significant relationships. The dashed arrow denotes a non-statistically significant relationship. This diagram depicts an observed set of associations and does not imply causality.

Visualizing the intention - action gap

The gap between stated intention and actual behavior is demonstrated in Fig. 2 below. Although most people claimed that their intention was positive to very positive, their actual frequency of donation was still low, occurring mostly in the baseline or "first time" category. The descriptive data combined with the regression analysis demonstrates that there is a discrepancy between the intention and behavior of young Indians: the former

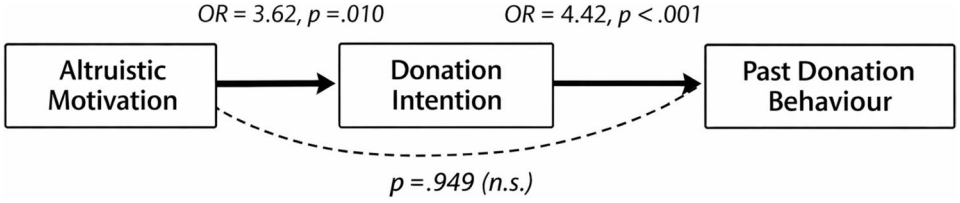


Fig. 1 Conceptual model showing the intention-action gap in blood donation. AM = Altruistic Motivation; DI = Donation Intention; PDB = Past Donation Behaviour; OR = Odds Ratio; p = Probability value; n.s. = Not significant. Conceptual representation of associations between altruistic motivation, donation intention, and past donation behavior

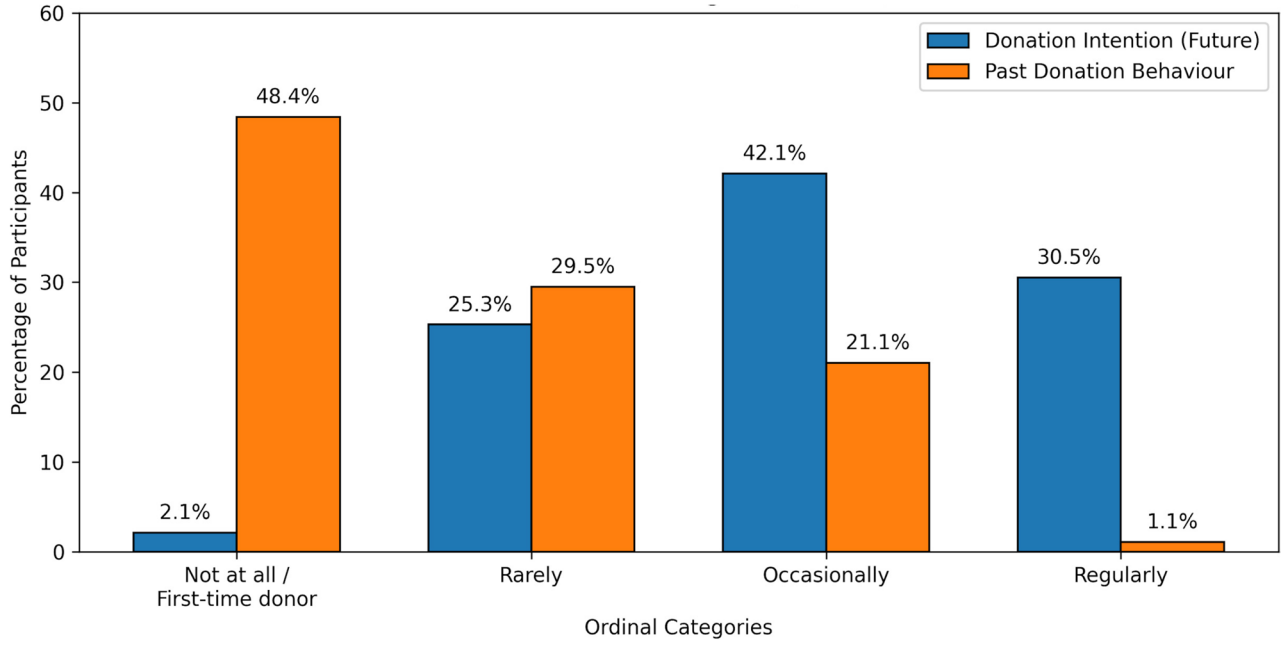


Fig. 2 Distribution of future donation intention and past donation behaviour across ordinal categories. Don. Int. = Donation Intention (Future); Past Don. Behav. = Past Donation Behaviour; Not at all/FTD = Not at all/First-time donor; Rarely = Rarely; Occasionally = Occasionally; Regularly = Regularly; N = Total sample size

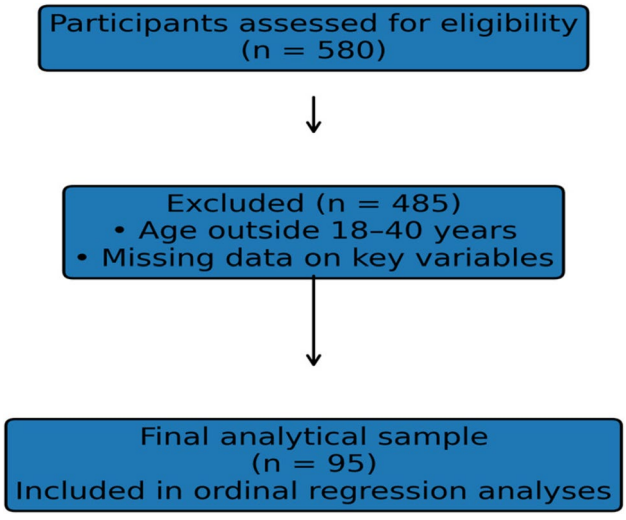


Fig. 3 STROBE flow diagram illustrating participant selection and analytical sample formation. Elig. = Assessed for eligibility; Excl. = Excluded; Age crit. = Age outside 18 - 40 years; Miss. data = Missing data on key variables; Final sample = Final analytical sample; Ord. reg. = Ordinal regression analyses; N = Total sample size

seems to be sufficiently driven by moral motives, while the latter isn't.

The selection of participants and exclusions are presented in a flow diagram that is consistent with STROBE guidelines (Fig. 3).

Discussion

The result of the current study revealed that there is a contradiction between intention and donation behaviors in the young Indians [10, 14]. Although altruism showed its positive influence on the intention to donate, it did not show such an effect when it comes to donation behaviors. These findings reflect the intention-behavior gap even though the data are cross-sectional in nature.

A key methodological limitation is that intention and past donation behaviour were measured at the same time. Therefore, the analysis cannot determine whether intention preceded behaviour, whether prior behaviour shaped current intention, or whether both were influenced by unmeasured factors. The findings should therefore be interpreted as cross-sectional associations rather than evidence of behavioral transition over time.

Practical implications must be taken with caution because current research does not address the barriers to blood donation such as convenience, fear, distance from donation centers, quality of service offered, and previous deferral experience. Therefore, recommendations regarding accessibility, humanized messaging, and donor-support strategies are informed by the broader blood donation literature in conjunction with the present findings, rather than arising solely from variables measured in this dataset.

Theoretical implications: beyond abstract altruism

This finding is in accordance with the Theory of Planned Behavior (TPB), according to which, intent represents the

closest measure for predicting behavior. But the discrepancy between higher levels of altruism and less frequent behavior indicates that while abstract moral principles are important, they may lack the motivational power needed to surpass practical constraints [10, 15].

As explained by [9], the association between altruism and behavior tends to follow the indirect route of humanization of the other party. In this regard, more general messages about altruism such as saving lives tend not to be sufficient in motivating people to take action. Humanizing the recipient can help to minimize anxiety associated with being part of different groups and generate empathy, leading to behavior [6, 8]. For recruitment agencies in India, this implies a shift from broad humanitarian slogans toward communication strategies that emphasize the humanized impact of donation.

Methodological considerations and "circularity"

One major drawback in terms of design in studies involving donations is the possible circularity involved when using past behavior to forecast intention. This is the case since we recognize that past behavior will serve to strengthen intention just as well as intention will forecast future behavior. However, a longitudinal study would be required in order to determine a clear timeline, but our "snap shot" study is important because it shows that there are a lot of people who exhibit high intention but do not have a track record of donation behavior. This reinforces the conclusion that high intention is a necessary, but not sufficient, condition for behavioral regularities.

Contextual barriers in the Indian context

The reduced intentions of donating blood amongst female participants have to be examined in the context of the health environment in India [2]. Medical disqualification due to low hemoglobin level is one of the main structural barriers that make women unwilling to create any sort of donation intention. Additionally, variables related to logistics such as physical proximity of participants to blood banks and ease of accessing mobile camps were not controlled for in the current analysis [16, 17]. It can be assumed that these factors might moderate the relationship between motivation and actual action regardless of any intention. In terms of recommendations based on the current findings, one has to remember that they do not imply any structural or logistic barriers to donations. Recommendations are based on prior research on blood donations and are not a result of direct testing performed within the current data set.

Female participants had lower odds of reporting higher donation intention in the present model. This association should be interpreted cautiously because the study did not measure hemoglobin status, medical deferral history, caregiving responsibilities, travel safety, or access to gender-sensitive donation facilities. These factors may be relevant in the Indian context, but they were not directly tested in this analysis.

These explanations remain speculative because hemoglobin status, prior deferral, caregiving responsibilities, travel safety, and access to donation sites were not measured in the present study.

Strengths and limitations

One of the most critical constraints is associated with the considerable number of people who have been excluded from the analysis. While those individuals were filtered using a certain criterion of age, completeness, and quality of the data provided, the number of respondents who had to be excluded turned out to be quite large. Thus, the remaining part of the sample is expected to differ from the overall young adult population. This limits confidence in the representativeness and stability of the observed associations. The findings should therefore be interpreted as exploratory and sample-specific rather than as population-level estimates.

The regression models did not include several potentially important confounding variables, including health eligibility, hemoglobin status, prior medical deferral, proximity to donation centres, fear of needles, prior adverse donation experiences, or frequency of exposure to donation camps. These unmeasured factors may partly account for the associations observed in the present analysis.

The study also has a conceptual limitation arising from the mismatch between future-oriented intention and retrospective self-reported behaviour. Because intention and past behaviour were measured at the same time, the analysis does not test whether intention translated into future donation behaviour. It only assesses whether current intention is associated with reported prior donation frequency.

Conclusion and practical recommendations

This study examined associations among altruistic motivation, donation intention, and self-reported past donation behaviour among young Indian adults. Altruistic motivation was associated with higher donation intention but was not independently associated with reported past donation frequency after intention was included in the model. Higher donation intention was associated with higher reported donation frequency, although the cross-sectional design prevents conclusions about temporal ordering or causal direction.

The findings suggest that motivation and reported behavioral engagement should be distinguished in blood donation research. However, given the small analytical sample, substantial exclusion rate, and reliance on self-reported data, the results should be interpreted as

exploratory. Longitudinal studies with larger and more representative samples are needed to clarify how donation intention relates to sustained future donation behaviour.

Institutional review board statement
The study was conducted in accordance with the Declaration of Helsinki and was approved by an Institutional Review Board/Ethics committee. See details under Methods.

Authors' contributions
Dr. Alaka Chandak was the Principal Investigator, conceptualized the study, and led the entire research. Dr. Girija Mahale provided an expertise emotional well-being point of view and helped in the theoretical perspective. Dr. Ketaki Data collection and manuscript drafting were carried out by Washikar, whereas the clinical expertise on the part of blood bank operation was given by Dr. Rajeshwari Kalbag. Dr. Manesh Muraleedharan contributed to study design, methodology, and data analysis. All authors collaborated with, reviewed, and approved the final version of the manuscript.

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Data availability
The data that support the findings of this study are available upon reasonable request: a written request can be submitted to investigator Dr. Alaka Chandak [dralaka@sihs.edu.in].

Declarations

Ethics approval and consent to participate
The protocol of this study was approved by the institutional ethics review board: Institutional Ethics Committee (IEC) approval was obtained from the committee of Symbiosis International (Deemed University) Approval no. is SIU/IEC/648. All the participants gave informed consent and responses were anonymized.

Competing interests
The authors declare no competing interests.

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